

# Modular Battery Management System (BMS-X800)

Safety, Environmental, and Multimodal Verification Requirements Matrix

| Property             | Specification Value                        | Property         | Specification Value            |
|----------------------|--------------------------------------------|------------------|--------------------------------|
| Product Category:    | Automotive High-Voltage Energy Storage     | Target Standard: | ISO 26262:2018 / FMVSS 305     |
| System Architecture: | 800V DC Nominal / 12 Cell Modules          | Integrity Level: | ASIL C / D Safety Critical     |
| Author:              | Apex EV Safety & Systems Architecture Team | Approval Status: | Released for Verification (Q3) |

## Section 1: High-Voltage Safety & Electrical Protection Requirements

REQ-BMS-001: High-Voltage Bus Active Discharge Timing

[Category: Electrical / Safety] [ASIL D]

Following an emergency shutdown trigger or vehicle collision event signal, the BMS active discharge circuit shall rapidly de-energize the high-voltage DC bus. The DC bus voltage ( $V_{bus}$ ) shall drop from its nominal operating potential (400.0 V) to below 60.0 V within a maximum allowable duration of 5.00 seconds, in strict compliance with FMVSS 305 Section S5.4.3.4 and ISO 6469-3.

REQ-BMS-002: Galvanic Isolation Resistance Monitoring

[Category: Electrical] [ASIL C]

Under all nominal DC operating conditions, the galvanic isolation resistance between the isolated high-voltage conductors (both positive rail DC+ and negative rail DC-) and the vehicle chassis electrical ground shall remain at or above 500.0 Ohm/V (total resistance  $\geq 200$  kOhm for 400V system). If isolation falls below this threshold, an electrical fault warning shall be raised within 1.0 second.

REQ-BMS-003: Over-Voltage Cell Contactor Disconnect Reaction Time

[Category: Electrical / Protection] [ASIL D]

If any individual cell sensing channel detects a terminal voltage exceeding 4.250 V for longer than 2.0 ms, the BMS hardware safety supervisor shall de-energize the main contactor coil driver. Total disconnection time from fault threshold detection to physical contactor air-gap separation shall not exceed 20.0 ms to prevent irreversible thermal and electrolytic cell degradation.

Section 2: Thermal Management & Environmental Operational Limits

REQ-BMS-004: Fast-Charge Cell-to-Cell Temperature Uniformity [Category: Thermal Management] [ASIL B]

During maximum continuous 3C DC fast-charging (from 10% to 80% State-of-Charge at peak 120 kW input), the active liquid cooling subsystem shall maintain thermal equilibrium across all module cells. The maximum temperature differential (Delta T) between any two cell temperature monitoring sensors throughout the pack shall not exceed 5.0 deg C at any point during the charging cycle.

REQ-BMS-005: Extreme Cold Ambient Operational Temperature Range [Category: Environmental] [ASIL B]

The battery system and control electronics shall maintain continuous operational telemetry, pre-charge capability, and safety monitoring across the full ambient climatic operating range from -40.0 deg C to +85.0 deg C. The system shall successfully execute pre-heating protocols and start communication without hardware fault or watchdog reset at -40.0 deg C cold-soak equilibrium.

REQ-BMS-006: Thermal Runaway Warning Broadcast Latency [Category: Thermal / Safety] [ASIL D]

Upon detection of venting gas (via combustible sensor) or localized temperature gradient exceeding 1.0 deg C/s, the BMS shall transmit an emergency thermal runaway warning frame (CAN-FD ID 0x018) to the vehicle master controller. The total latency from sensor threshold trip to bus frame transmission shall be strictly less than 100 ms to ensure sufficient egress time for vehicle occupants.

Section 3: Mechanical Enclosure, Visual Safety & Crash Protection

REQ-BMS-007: High-Voltage Enclosure Hazard Warning Labeling [Category: Mechanical / Visual] [Visual]

The exterior surface of the battery top cover shall bear a permanent, weather-resistant ISO 7010-W012 high-voltage hazard warning triangle symbol. The warning symbol shall have an equilateral triangle height of at least 40.0 mm and shall be positioned directly adjacent to the Manual Service Disconnect (MSD) port. Compliance shall be verified by physical photographic measurement and dimensional inspection.

REQ-BMS-008: High-Voltage Terminal Creepage Distance to Chassis [Category: Physical / Clearance] [Visual]

The shortest physical creepage path along the surface of insulating barriers between any uninsulated live high-voltage terminal busbar and the grounded metal enclosure housing shall be at least 8.00 mm. Compliance shall be verified through optical CMM cross-sectional dimensional verification under ISO 6469-3 Pollution Degree 2 environmental conditions.

REQ-BMS-009: Submersion Ingress Protection (IP67) [Category: Ingress Protection] [ASIL C]

The fully sealed battery enclosure assembly shall satisfy IP67 protection per ISO 20653. When submerged under a hydrostatic water column depth of 1.00 meter for a continuous duration of 30.0 minutes, there shall be exactly 0.0 mL of liquid water ingress into the internal high-voltage chamber, and internal cavity pressure shall remain hermetically stabilized.

REQ-BMS-010: Pyrotechnic Disconnect Trigger on Crash Deceleration [Category: Mechanical / Crash] [ASIL D]

In the event of a severe vehicle collision where deceleration along the longitudinal X-axis exceeds 15.0 g for a duration greater than 10.0 ms, the BMS crash sensing subsystem shall issue a pyrotechnic ignition pulse to detonate the pyrofuse switch within 5.0 ms of acceleration threshold confirmation, physically severing the battery traction power circuit.